

Individual Reflection on Group Project

Project Title: Edge Deployable Rock Detection and Burial Status Classification

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Team / Group: Team 9

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1. Your views on the completing the project as a team

Working as a team was valuable because the project combined dataset preparation, detector development, experimental research and edge deployment. My primary work involved training, validating and testing the YOLO11s and creating the CLIP ViT-B/32 branch for burial status classification. YOLO11s helped in the controlled comparison with YOLOv8n and YOLO11n, and CLIP took the approach of tackling things from the crop level. When multiple people are contributing to one experimental result, it is important to have a shared set of evaluation rules and to have clear separation between experimental branches.

2. Your views on project management and execution of your team

The team trained and tested all three YOLO architectures in a similar way to ensure a fair comparison. YOLO11s excelled in recall, F1 score, and mAP50, whereas the predefined metric of mAP50-95 helped to choose YOLOv8n. In parallel, I used the CLIP zero-shot classification pipeline, obtained frozen image embeddings, trained a linear probe and experimented with a YOLO-CLIP crop-verification pipeline. This separation of CLIP from the Raspberry Pi benchmark prevented the mixing of different evaluation units and second model cost, which is unmeasured. This highlighted for me the need of establishing experimental limits in advance and documenting why a promising branch is retained or excluded.

3. Reflections on Successes, Failures, and Retrospective Changes

- **What Worked Well:** YOLO11s achieved validation recall of 0.801, F1 of 0.782 and mAP50 of 0.783. Compared to zero-shot CLIP, the CLIP linear probe boosted the test accuracy by 47.4% to 89.6% and macro F1 by 0.450 to 0.895, showcasing the power of task-specific supervision.
- **What Did Not Work Well:** YOLO11s did not score the best mAP50-95 and was significantly larger than the nano models. The already trained CLIP branch couldn't be included, as there was no verified Raspberry Pi benchmark, and Zero-shot CLIP was unreliable.
- **What I Would Do Differently:** Later on I would design the CLIP evaluation plan and more testing of prompt and backbone variants; I would also employ stronger repeated validation. If time permitted I would also test the YOLO-CLIP pipeline with the crops predicted, and determine its latency and memory gain on the Raspberry Pi.

4. Any other reflections

This project helped me to improve my skills in making comparisons between models without exaggerating their metrics. I got to know that having multiple secondary measures does not invalidate the selection criterion being declared and that accuracy of crop classification cannot be compared directly to the object-detection mAP. Although the CLIP branch was not used in the final Raspberry Pi benchmark, it was still useful as a potential future verification stage. These lessons will impact my design, evaluation, and communication of future machine-learning experiments.